

Supplementary materials

Comparing our standalone methods

**Comparing our standalone methods against
DLX⁺: LP Boost**

**Optimality gap and solution time comparisons
among all methods**

**Comparing DLX⁺ against Gurobi
DLX⁺ flowchart**

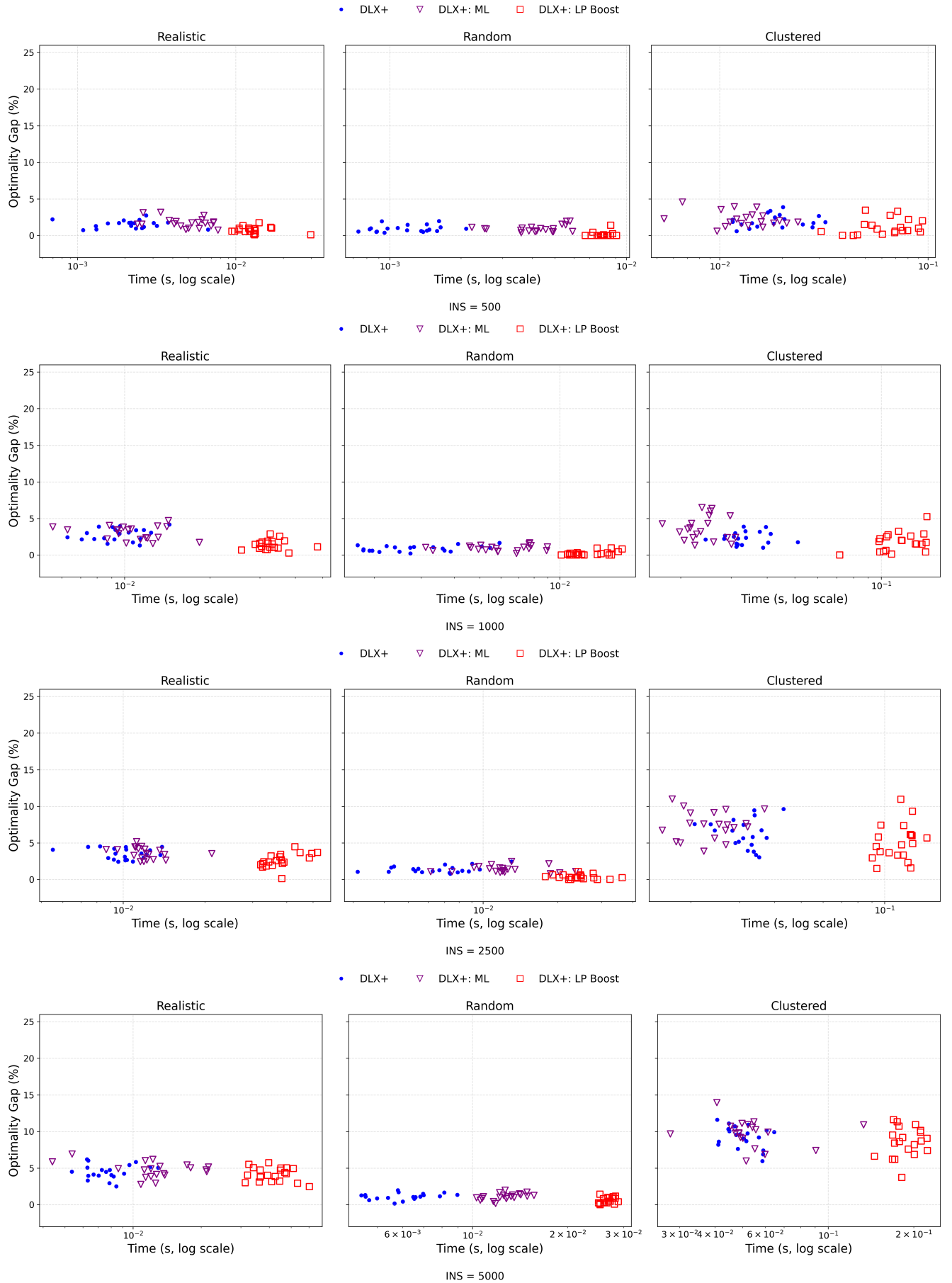


Figure 1: Optimality gap versus computation time in seconds on a logarithmic scale across different user distributions (Realistic, Random, Clustered) for user sizes (INS = 500, 1000, 2500, 5000), comparing DLX+, DLX+: ML and DLX+: LP Boost.

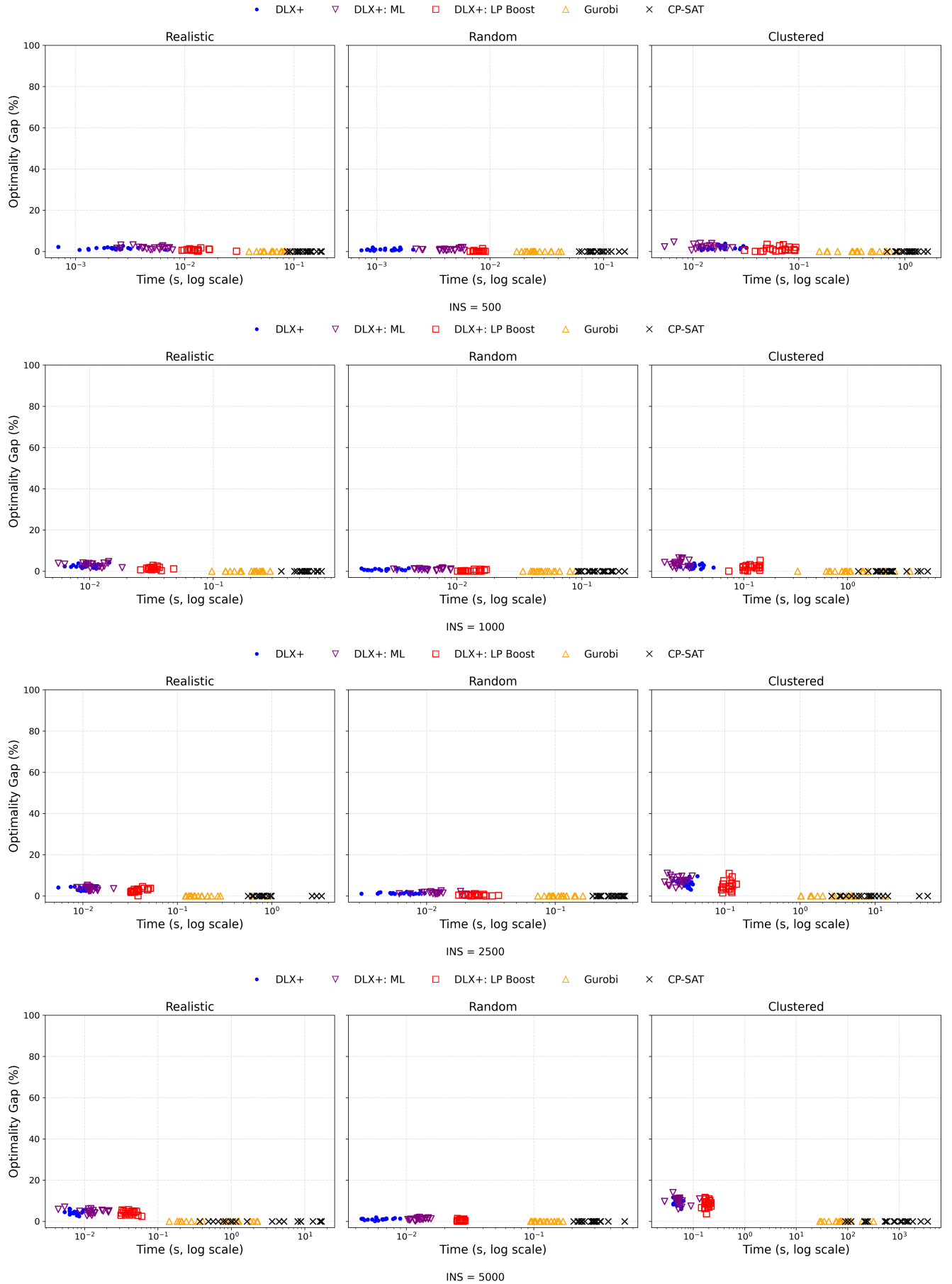
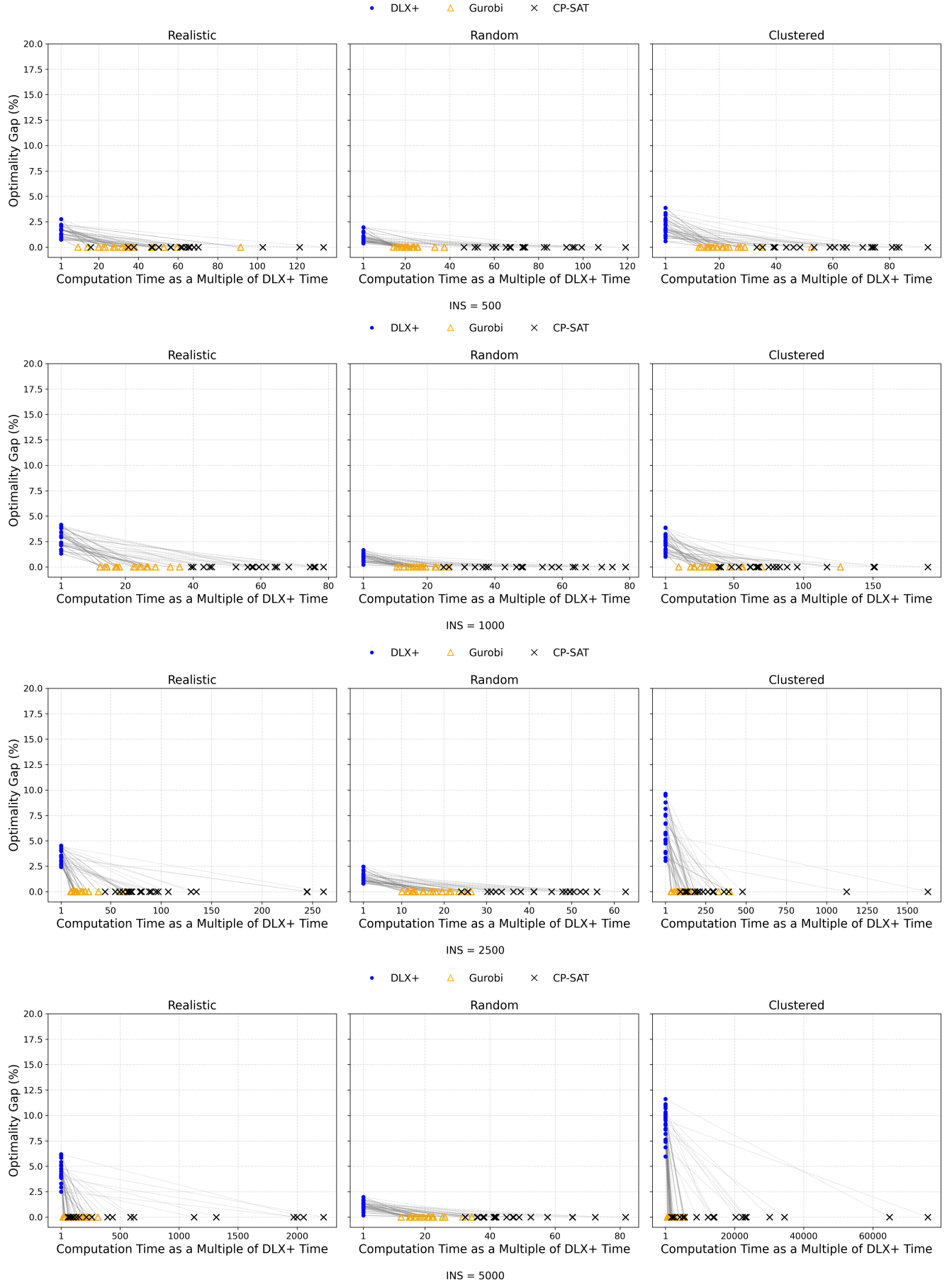


Figure 2: Optimality gap versus computation time in seconds on a logarithmic scale across different user distributions (Realistic, Random, Clustered) for user sizes ($INS = 500, 1000, 2500, 5000$), comparing DLX⁺, DLX⁺: ML, DLX⁺: LP Boost, Gurobi and CP-SAT.



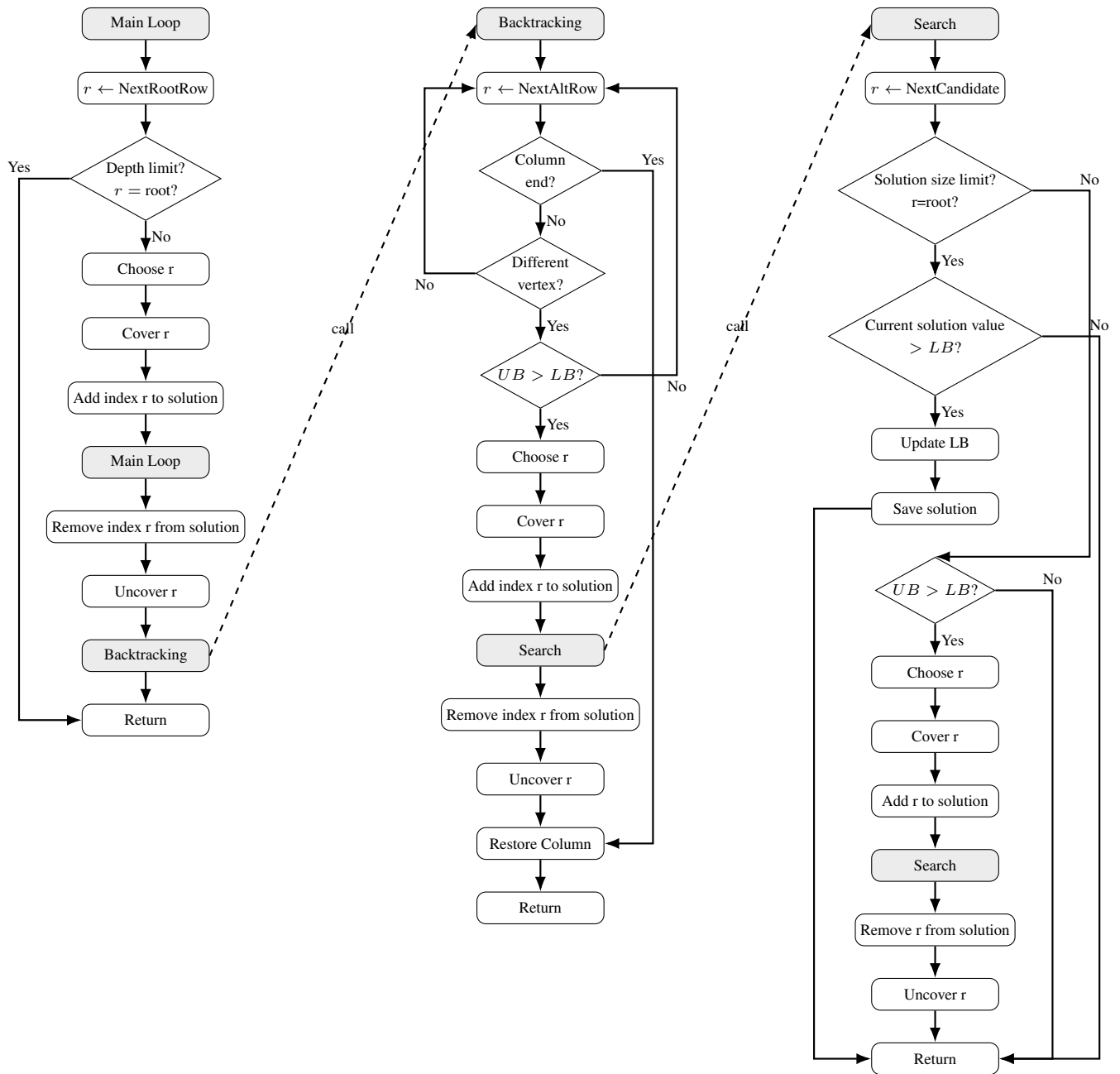


Figure 4: Control flow of the DLX+ main loop, backtracking, and search procedures. Dashed arrows indicate procedure calls.